

$A \geq B$: An LCF-Style Sum-of-Squares Prover for Polynomial Inequalities

Mathematical Foundations and System Design

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Abstract

$A \geq B$ is a standalone system that proves polynomial inequalities of the form $A \geq B$ by rewriting them as $p = A - B \geq 0$ and certifying $p \geq 0$ with a *sum-of-squares* (SOS) decomposition $p = \sum_i c_i q_i^2$ with $c_i \in \mathbb{Q}_{\geq 0}$ and $q_i \in \mathbb{Q}[x_1, \dots, x_n]$. The system is built in the LCF tradition: an *untrusted* search engine proposes certificates, and a small, *trusted*, exact-rational *checker* is the sole authority on whether an inequality is proved. This document develops the mathematics from first principles—nonnegativity, sum-of-squares certificates and their soundness, the Gram-matrix / semidefinite reformulation, and exact rational extraction via $LDL^\top$ —and then describes the concrete system: exact representations, the trusted checker, the prover, denominator clearing, the validation harness, and the road to a formally verified checker in Lean 4.

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1 Introduction

Deciding whether a multivariate polynomial is nonnegative on $\mathbb{R}^n$ is, in general, NP-hard. A practical and widely used relaxation restricts attention to polynomials that admit a *certificate* of nonnegativity in the form of a sum of squares. Such certificates are attractive for two reasons. First, they are *sound by inspection*: a sum of squares (with nonnegative coefficients) is manifestly nonnegative. Second, searching for them reduces to *semidefinite programming*, a tractable convex problem.

$A \geq B$ applies this idea to the concrete task of proving inequalities such as

$$a^2 + b^2 + c^2 \geq ab + bc + ca,$$

by exhibiting and verifying an explicit algebraic identity, here

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}(a - b)^2 + \frac{1}{2}(b - c)^2 + \frac{1}{2}(a - c)^2,$$

whose right-hand side is visibly nonnegative.

The organising principle is the separation, due to the LCF family of proof assistants, between an *untrusted engine* and a *trusted kernel*. The engine may use any heuristic, numerical routine, or approximate solver; whatever it produces is re-verified, exactly, by a small checker. The program never reports PROVED unless the checker accepts. Consequently a bug in the (complicated) search can only cause a *missed* proof, never a false one. Section 2 develops the mathematics; Sections 3–6 describe the system; Section 7 gives the architecture; and Section 9 sketches the remaining work, including formal verification of the checker in Lean 4.

2 Mathematical foundations

2.1 Polynomials and nonnegativity

Fix variables $\mathbf{x} = (x_1, \dots, x_n)$. We work in the polynomial ring $\mathbb{Q}[\mathbf{x}]$ over the rationals. A *monomial* is a product $\mathbf{x}^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ indexed by an exponent vector $\alpha \in \mathbb{N}^n$; its *degree* is $|\alpha| = \sum_i \alpha_i$. A polynomial $p = \sum_\alpha c_\alpha \mathbf{x}^\alpha$ has finitely many nonzero coefficients $c_\alpha \in \mathbb{Q}$, and $\deg p = \max\{|\alpha| : c_\alpha \neq 0\}$. Every polynomial defines a function $\mathbb{R}^n \rightarrow \mathbb{R}$ by evaluation.

Definition 2.1 (Nonnegative polynomial). $p \in \mathbb{Q}[\mathbf{x}]$ is *nonnegative*, written $p \geq 0$, if $p(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.

2.2 Sum-of-squares certificates

Definition 2.2 (Sum of squares). $p \in \mathbb{Q}[\mathbf{x}]$ is a *sum of squares* (SOS) if there exist rationals $c_1, \dots, c_m \geq 0$ and polynomials $q_1, \dots, q_m \in \mathbb{Q}[\mathbf{x}]$ with

$$p = \sum_{i=1}^m c_i q_i^2. \tag{1}$$

We call the list $((c_i, q_i))_{i=1}^m$ an *SOS certificate* for p .

The whole system rests on the following elementary but decisive fact.

Theorem 2.3 (Soundness of SOS certificates). *If p admits an SOS certificate (1), then $p \geq 0$.*

Proof. For every $\mathbf{x} \in \mathbb{R}^n$ and each i , $q_i(\mathbf{x})^2 \geq 0$ because it is the square of a real number, and $c_i \geq 0$; hence $c_i q_i(\mathbf{x})^2 \geq 0$. A finite sum of nonnegative reals is nonnegative, so $p(\mathbf{x}) = \sum_i c_i q_i(\mathbf{x})^2 \geq 0$. $\square$

Theorem 2.3 is an *implication*. Its converse is false: not every nonnegative polynomial is a sum of squares.

Example 2.4 (Motzkin polynomial). The polynomial

$$M(x, y) = x^4 y^2 + x^2 y^4 + 1 - 3x^2 y^2$$

satisfies $M \geq 0$ on $\mathbb{R}^2$ (by the AM–GM inequality applied to $x^4 y^2$, $x^2 y^4$, and 1), yet M is *not* a sum of squares. It is the classical witness that nonnegativity is strictly weaker than the SOS property.

The exact boundary is Hilbert’s theorem.

Theorem 2.5 (Hilbert, 1888). *Every nonnegative real form (homogeneous polynomial) of degree $2d$ in n variables is a sum of squares if and only if $n \leq 2$, or $2d = 2$, or $(n, 2d) = (3, 4)$. In all other cases there exist nonnegative forms that are not sums of squares.*

Two consequences shape the design.

- **Degree ≤ 2 is complete.** For quadratic p , nonnegativity and the SOS property coincide (Proposition 5.2); the search can never fail on a true quadratic inequality.
- **Higher degree is incomplete.** From degree 6 in 3 variables (Example 2.4) onward, some true inequalities have no SOS certificate. The prover must therefore report a third outcome, `NO_CERT_FOUND`, which is *not* a disproof.

2.3 The reduction

An inequality $A \geq B$ with $A, B \in \mathbb{Q}[\mathbf{x}]$ is equivalent to $A - B \geq 0$. Setting $p := A - B$ (or $p := B - A$ for $A \leq B$) reduces every claim to the single question: *does p admit an SOS certificate?* The rest of the paper answers this constructively and verifies the answer exactly.

3 Exact representation

Because the checker must decide the polynomial identity (1) *exactly*, every object is represented over $\mathbb{Q}$ with no floating point.

Rationals.

Arbitrary-precision $\mathbb{Q}$ (via a bignum library). All coefficient arithmetic is exact; sign tests and equality are decidable.

Monomials.

Exponent vectors, e.g. $a^2 b c^3 \mapsto [2, 1, 3]$. The canonical form strips trailing zeros, so each monomial has a unique representation ($a^2 \mapsto [2]$ regardless of the ambient variable count), and the constant monomial is the empty vector.

Polynomials.

A finite map from canonical monomials to *nonzero* rational coefficients. Maintaining this normal form (no stored zeros, like terms combined) makes polynomial equality *structural*: $p = q$ iff their maps are identical.

Remark 3.1. Structural equality of normal forms is what makes the trusted core small: the checker never reasons about polynomials symbolically; it computes both sides of (1) into normal form and compares maps.

4 The trusted checker

The checker is the only component that must be correct. Given a target p and a candidate certificate $((c_i, q_i))$ it performs the three exact steps in Algorithm 1.

```

check_sos (p, cert):
  if any c_i < 0 in cert      return false    # (1) nonnegative coefficients
  R := sum_i c_i * (q_i * q_i)              # (2) exact rational expansion
  return (p == R)                      # (3) equality of normal forms

```

Figure 1: The trusted checker `check_sos`.

Proposition 4.1 (Checker soundness). *If `check_sos`(p, cert) returns `true`, then $p \geq 0$.*

Proof. On returning `true`, step (1) guarantees $c_i \geq 0$ for all i , and steps (2)–(3) guarantee the polynomial identity $p = \sum_i c_i q_i^2$ holds in $\mathbb{Q}[\mathbf{x}]$ (equal normal forms are equal polynomials). Theorem 2.3 then gives $p \geq 0$. $\square$

Proposition 4.1 is the entire trusted-computing-base argument: it depends only on exact rational arithmetic, exact polynomial multiplication and addition, and structural equality—nothing about how the certificate was found.

5 The prover

The prover is the untrusted search for a certificate. Its mathematical core is the Gram-matrix (or “Gram-matrix method”) reformulation of the SOS condition.

5.1 The Gram-matrix reformulation

Fix a finite vector of monomials $\mathbf{z} = (z_1, \dots, z_r)^\top$, the *basis*. For a symmetric matrix $Q \in \mathbb{Q}^{r \times r}$ consider the quadratic form $\mathbf{z}^\top Q \mathbf{z} = \sum_{i,j} Q_{ij} z_i z_j$, a polynomial.

Lemma 5.1 (Gram representation). *Let $\mathbf{z}$ be a monomial basis. A polynomial p is a sum of squares of polynomials in $\text{span } \mathbf{z}$ if and only if there exists a symmetric positive semidefinite matrix $Q \succeq 0$ with $p = \mathbf{z}^\top Q \mathbf{z}$.*

Proof. ($\Leftarrow$) If $Q \succeq 0$, decompose $Q = \sum_k \lambda_k \mathbf{v}_k \mathbf{v}_k^\top$ with $\lambda_k \geq 0$ (e.g. a spectral decomposition). Then

$$\mathbf{z}^\top Q \mathbf{z} = \sum_k \lambda_k (\mathbf{v}_k^\top \mathbf{z}) (\mathbf{z}^\top \mathbf{v}_k) = \sum_k \lambda_k (\mathbf{v}_k^\top \mathbf{z})^2,$$

an SOS with square bases $q_k = \mathbf{v}_k^\top \mathbf{z} \in \text{span } \mathbf{z}$.

($\Rightarrow$) If $p = \sum_k \lambda_k q_k^2$ with $\lambda_k \geq 0$ and $q_k = \mathbf{v}_k^\top \mathbf{z}$, set $Q = \sum_k \lambda_k \mathbf{v}_k \mathbf{v}_k^\top \succeq 0$; then $\mathbf{z}^\top Q \mathbf{z} = \sum_k \lambda_k q_k^2 = p$. $\square$

Matching coefficients turns $p = \mathbf{z}^\top Q \mathbf{z}$ into *linear* constraints on Q : grouping the products $z_i z_j$ by the monomial μ they equal,

$$\sum_{(i,j): z_i z_j = \mu} Q_{ij} = \text{coeff}_p(\mu) \quad \text{for every monomial } \mu. \quad (2)$$

The solution set of (2) is an affine subspace of symmetric matrices; finding a positive semidefinite point in it is precisely a *semidefinite feasibility problem*. Two regimes arise, treated next.

5.2 Degree two: the unique Gram matrix

Take the basis $\mathbf{z} = (1, x_1, \dots, x_n)^\top$. Every monomial of degree ≤ 2 is the product of *exactly one* unordered pair of basis elements: $1 = 1 \cdot 1$, $x_i = 1 \cdot x_i$, $x_i^2 = x_i \cdot x_i$, and $x_i x_j = x_i \cdot x_j$. Hence (2) determines Q uniquely:

$$Q_{00} = \text{coeff}_p(1), \quad Q_{0i} = \frac{1}{2} \text{coeff}_p(x_i), \quad Q_{ii} = \text{coeff}_p(x_i^2), \quad Q_{ij} = \frac{1}{2} \text{coeff}_p(x_i x_j) \quad (i < j).$$

Proposition 5.2 (No gap in degree two). *Write $p(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} + 2 \mathbf{b}^\top \mathbf{x} + c$ and let $Q = \begin{pmatrix} c & \mathbf{b}^\top \\ \mathbf{b} & A \end{pmatrix}$ be the Gram matrix above. Then the following are equivalent: (i) $p \geq 0$; (ii) $Q \succeq 0$; (iii) p is a sum of squares.*

Proof. (ii) $\Rightarrow$ (iii) is Lemma 5.1, and (iii) $\Rightarrow$ (i) is Theorem 2.3. For (i) $\Rightarrow$ (ii): for any $\mathbf{x}$ and $t \neq 0$, $(\mathbf{x}, t) Q (\mathbf{x}, t)^\top = t^2 p(\mathbf{x}/t) \geq 0$; taking $t \rightarrow 0$ and using continuity gives $(\mathbf{x}, t) Q (\mathbf{x}, t)^\top \geq 0$ for all $(\mathbf{x}, t)$, i.e. $Q \succeq 0$. $\square$

Thus for quadratic targets the prover is *complete*: it succeeds exactly when the inequality is true.

5.3 Exact extraction: rational $LDL^\top$

It remains to turn a rational PSD Q into an explicit rational SOS. This is done by symmetric Gaussian elimination.

Lemma 5.3 (Zero pivots of a PSD matrix). *If $Q \succeq 0$ and $Q_{kk} = 0$, then $Q_{kj} = 0$ for all j .*

Proof. The principal 2×2 submatrix on indices $\{k, j\}$ is $\begin{pmatrix} 0 & Q_{kj} \\ Q_{kj} & Q_{jj} \end{pmatrix}$ and must be PSD; its determinant $-Q_{kj}^2 \geq 0$ forces $Q_{kj} = 0$. $\square$

Theorem 5.4 (Rational PSD $\Rightarrow$ rational SOS). *Let $Q \in \mathbb{Q}^{r \times r}$ be symmetric and PSD. The factorisation $Q = LDL^\top$ with L unit lower-triangular and D diagonal, computed by*

$$D_k = Q_{kk} - \sum_{j < k} L_{kj}^2 D_j, \quad L_{ik} = \frac{1}{D_k} \left(Q_{ik} - \sum_{j < k} L_{ij} L_{kj} D_j \right) \quad (i > k), \quad (3)$$

has $L, D \in \mathbb{Q}$ and $D_k \geq 0$ for all k . Consequently

$$p = \mathbf{z}^\top Q \mathbf{z} = \sum_{k: D_k > 0} D_k q_k^2, \quad q_k = (L^\top \mathbf{z})_k = \sum_j L_{jk} z_j \in \mathbb{Q}[\mathbf{x}],$$

is an exact rational SOS certificate for p .

Proof. The quantity D_k is the k -th pivot, equal to the Schur complement diagonal entry; for a PSD matrix all pivots are ≥ 0 . If $D_k > 0$, (3) is a well-defined rational division. If $D_k = 0$, then by Lemma 5.3 (applied to the Schur complement, itself PSD) the remaining entries of that column vanish, so no division occurs and $L_{ik} = 0$; no pivoting is needed. All operations are in $\mathbb{Q}$, so L, D are rational. Finally $Q = LDL^\top$ gives $\mathbf{z}^\top Q \mathbf{z} = (L^\top \mathbf{z})^\top D (L^\top \mathbf{z}) = \sum_k D_k (L^\top \mathbf{z})_k^2$, and the terms with $D_k = 0$ drop. $\square$

Corollary 5.5. *A symmetric rational positive semidefinite matrix is a sum of rational squares. Over $\mathbb{Q}$, for the Gram problem, “PSD” and “SOS-decomposable” coincide, and the decomposition is obtained exactly.*

Example 5.6 (Worked degree-two case). For $p = a^2 - ab + b^2$ and $\mathbf{z} = (a, b)^\top$, $Q = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$.

Then (3) gives $D_1 = 1$, $L_{21} = -\frac{1}{2}$, $D_2 = 1 - \frac{1}{4} = \frac{3}{4}$, so

$$a^2 - ab + b^2 = 1 \cdot \left(a - \frac{1}{2}b\right)^2 + \frac{3}{4}b^2.$$

5.4 Higher degree and the semidefinite regime

For $\deg p > 2$ one enlarges the basis $\mathbf{z}$ (for a homogeneous target of degree $2d$, the monomials of degree d ; more economically, the “square roots” of the perfect-square monomials occurring in p). Two situations occur.

Unique Gram. If every product monomial in (2) arises from a single pair, Q is again determined, and Theorem 5.4 finishes the job. For instance $a^4 + b^4 \geq 2a^2b^2$ uses $\mathbf{z} = (a^2, b^2)^\top$ and $Q = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, giving $(a^2 - b^2)^2$.

Under-determined Gram (an SDP). When a monomial is the product of several pairs—e.g. $a^2b^2 = (a^2)(b^2) = (ab)(ab)$ —the constraints (2) leave free parameters, and one must *choose* a PSD member of the affine family. This is a genuine semidefinite program. The current implementation performs a bounded rational grid search over up to two free entries; every candidate satisfies $\mathbf{z}^\top Q \mathbf{z} = p$ by construction, so any PSD candidate yields a valid certificate, and the trusted checker verifies it. The planned replacement solves the SDP numerically and rounds the result to an exact rational PSD matrix (Section 9). Targets with no SOS representation at all (Example 2.4) correctly yield NO_CERT_FOUND.

Soundness is independent of all of this. Regardless of which basis, which grid point, or which numerical solver produces Q , the certificate is re-checked by `check_sos`. The search is untrusted; only Proposition 4.1 is relied upon.

6 Clearing denominators

Many olympiad inequalities are stated with fractions. The parser accepts division between expressions, and the normaliser reduces a rational-function claim to a polynomial one, soundly.

Proposition 6.1 (Denominator clearing). *Let A, B be rational functions and write $A - B = N/D$ with $N, D \in \mathbb{Q}[\mathbf{x}]$ and $D \neq 0$. Then, at every point where $D \neq 0$,*

$$A - B \geq 0 \iff N \cdot D \geq 0.$$

Hence proving $p := N \cdot D \geq 0$ on all of $\mathbb{R}^n$ establishes $A \geq B$ wherever it is defined.

Proof. Where $D \neq 0$ we have $D^2 > 0$, and N/D and $N \cdot D = (N/D) D^2$ have the same sign. Thus $N/D \geq 0 \iff ND \geq 0$. The claim follows, and note that no assumption is made on the sign of D . $\square$

The construction proceeds by evaluating each subexpression to a pair (numerator, denominator) and combining fractions; when the input contains no division, $D \equiv 1$ and $p = A - B$ as before, so the polynomial case is recovered exactly.

7 System design

7.1 Pipeline and the trust boundary

Figure 2 shows the flow from an input string to a rendered proof. The defining invariant is the one-way valve at the checker: the status PROVED is emitted *iff* the trusted checker accepts.

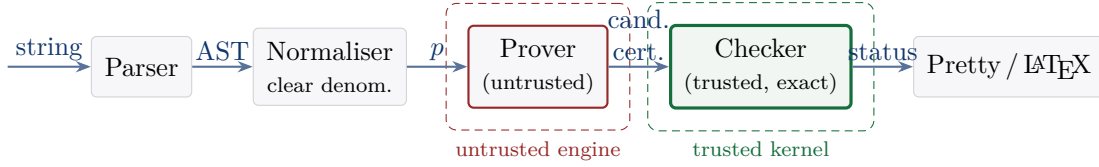


Figure 2: The $A \geq B$ pipeline. Only the checker is trusted; a fault in the prover can cause a missed proof, never a false one.

7.2 Module layering

The library is layered so that each module depends only on those below it, which keeps the trusted checker’s dependencies minimal:

Rational $\rightarrow$ Monomial $\rightarrow$ Polynomial $\rightarrow$ {Ast, Pretty} $\rightarrow$ {Parser, Normalizer, Certificate} $\rightarrow$ Checker $\rightarrow$ Prover.

Rational is the sole point of contact with the bignum backend, which localises the exact-arithmetic dependency (and eases a future pure-rational or formally verified reimplementation).

7.3 Command-line surface

The tool exposes `prove "<inequality>"` (parse, clear denominators, and search) and `check <cert.json>` (load and verify a certificate), reporting one of four statuses with matching exit codes: `PROVED`, `NO_CERT_FOUND`, `INVALID_INPUT`, `CHECK_FAILED`.

8 Validation

Correctness is defended empirically by a benchmark corpus of 119 inequalities, each tagged by category and by the status the system should return (Table 1). The corpus is drawn from classical results (AM–GM, Cauchy–Schwarz, Schur), named competition problems (IMO, APMO, and national olympiads), and a standard olympiad reference; it also includes deliberate soundness traps.

Category	Meaning	Expected status
<code>in_scope_sos</code>	nonnegative on $\mathbb{R}^n$, SOS over $\mathbb{Q}$	<code>PROVED</code>
<code>out_of_scope_v1</code>	true only under side constraints	<code>NO_CERT_FOUND</code>
<code>not_sos</code>	nonnegative but not SOS (Motzkin, Choi–Lam)	<code>NO_CERT_FOUND</code>
<code>false</code>	deliberately false	<code>NO_CERT_FOUND</code>

Table 1: Corpus categories and the status each should elicit.

Three independent harnesses guard the corpus: one runs every SOS certificate through the trusted checker; one is an *independent* numeric oracle that random-samples each claim on its domain (implemented in a different language, so an error in the core cannot mask a bad entry); and one enforces the *soundness gate*—no `false`, `constrained`, or `not_sos` entry may ever be proved. The gate is not decorative: it flagged a problem originally recorded as requiring positivity, whose difference is in fact the perfect square

$$(a + b)^2(a + c)^2 - 4abc(a + b + c) = (a^2 + ab + ac - bc)^2,$$

prompting its reclassification as an unconditional SOS identity.

9 Roadmap

Four extensions complete the design; each preserves the checker as the sole authority.

1. **Wide semidefinite prover.** Replace the bounded grid search by a genuine SDP over the affine family (2): the exact core emits the Gram program, a numerical solver (e.g. an interior-point method) returns an approximate PSD Q , and the core rounds it to an exact rational PSD matrix and re-verifies. This is the natural home for numerical code, cleanly quarantined behind the trust boundary.
2. **Constrained inequalities.** Admit side constraints ($a \geq 0$, $abc = 1$, triangle conditions) and search for *Positivstellensatz* certificates $p = \sigma_0 + \sum_i \sigma_i g_i$ with each σ_i a sum of squares and g_i a constraint. This subsumes the bulk of olympiad inequalities.
3. **A formally verified checker in Lean 4.** Reimplement the checker over `MvPolynomial (Fin n) ℚ` and prove its soundness theorem,

$$\text{check_sos } p \text{ cert} = \text{true} \implies \forall \mathbf{x} : \text{Fin } n \rightarrow \mathbb{R}, 0 \leq p.\text{eval } \mathbf{x},$$

which is exactly Proposition 4.1 discharged mechanically. The proof follows the pattern of Theorem 2.3 (`sq_nonneg`, `mul_nonneg`, `Finset.sum_nonneg`) once `check_sos` establishes the identity. This upgrades the trusted core from “small and audited” to machine-verified. (Lean 4 compiles to native code, so the verified checker also runs.)

4. **Interface.** Because the engine is pure functional code, it can be compiled to run in a browser, exposing a small structured interface that renders the existing L^AT_EX proof output.

Together, items 1 and 3 realise a principled three-language architecture: exact symbolic computation and the trusted core in a typed functional language, numerical semidefinite programming (untrusted) in the scientific-computing ecosystem, and a machine-checked proof of the kernel in a proof assistant.

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